

Hebrew Acronym Disambiguation: Open Generation or Candidate-Constrained Selection?

NLP Final Project Proposal

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Problem and motivation. Hebrew acronyms and abbreviations, often written with gershayim or geresh, are common in modern Hebrew. Many forms are genuinely ambiguous: the same acronym can correspond to unrelated long forms, and the correct expansion depends on context and domain. We study whether this problem should be treated as open-ended generation or as candidate-constrained disambiguation: given a sentence or short passage with a marked acronym, should a system freely generate an expansion, or select one from a verified inventory? This distinction matters because unconstrained LLMs may produce plausible but invalid expansions, while candidate-constrained systems can trade expressive freedom for reliability, privacy, and low-cost bulk processing.

Prior work and gap. To further research in modern Hebrew NLP, Shmidko et al. released DictaBERT [3], establishing a strong foundation for Hebrew language modeling. Addressing abbreviations, HaCohen-Kerner et al. presented HAADS [2], a system for disambiguation in Jewish-Law documents. However, modern acronym disambiguation is benchmark-driven and mostly English-focused. For example, Veyseh et al. introduce SciAD [1], a massive English dataset, with no equivalent resource available in Hebrew. These works motivate our formulation, but do not provide a public benchmark for *ambiguous Hebrew acronyms in context*, nor a controlled comparison between open LLM generation and candidate-constrained Hebrew disambiguation.

Data and scope. We construct a controlled benchmark of ambiguous Hebrew acronyms in context, with separate train, development, and held-out evaluation splits. The data source is Hebrew Wikipedia: article text plus abbreviation and disambiguation pages used to seed a manually normalized acronym-to-expansions inventory. We manually curate 30–50 acronym types, keeping only forms with at least two plausible expansions. Weakly labeled train/dev examples (1,000–2,000 instances) are mined by replacing full long-form candidates in context with the acronym. The held-out set (300–500 instances) is manually verified and may include naturally occurring acronym mentions if the context uniquely supports one expansion. Splits are by source document, with an optional type-held-out stress test, to reduce leakage.

Research question. Our research question is: *Which formulation is more reliable for ambiguous Hebrew acronym expansion: open-ended generation or candidate-constrained selection?* We answer it by building a controlled Hebrew benchmark and comparing unconstrained LLM prompting, candidate-list LLM prompting, frequency and classical baselines, and a small Hebrew encoder under the same verified candidate inventory.

Method and baselines. Our model arm is a candidate-constrained Hebrew ranker: given a marked context and a proposed expansion, a small Hebrew encoder scores whether the candidate is correct. The core project, however, is the benchmark and controlled comparison between generation and selection. All methods are evaluated on the same benchmark and candidate inventory. The default backbone is DictaBERT [3]. We fine-tune a lightweight classification head or parameter-efficient adapter over context-candidate pairs, using other candidate expansions for the same acronym as negatives. Baselines will include:

- most-frequent expansion per acronym;
- a classical TF-IDF/context-window logistic-regression or SVM classifier, as a lightweight HAADS [2]-style baseline;
- unconstrained zero-/few-shot LLM expansion, measured also for invalid candidate hallucinations;
- candidate-list LLM forced choice, to make the comparison fair under the same candidate inventory.

Evaluation. Primary metrics will be micro and macro Top-1 accuracy, MRR/Top-3 accuracy, and invalid-expansion rate for free-generation LLM outputs (outputs outside the verified candidate inventory). We will report inventory coverage for natural mentions, plus breakdowns by acronym frequency, number of candidate expansions, source type (masked long-form vs. natural mention), and context evidence: explicit lexical cue/definition, broader topical cue, or little/no local evidence. We will tune on development data only, run several seeds for fine-tuning, and use multiple prompt templates for LLM baselines. If the trained model initially fails, we will first verify that it can overfit a small sample, as a sanity check.

Resources and fallback. All core experiments fit the course Slurm `studentkillable` GPUs: the model is BERT-sized and training is a small classifier/adaptation experiment, not pretraining. Model weights are public HuggingFace checkpoints; the data comes from Hebrew Wikipedia under its public license; the team will manually verify the held-out set, with no external annotators assumed. We may need modest API credits (estimated \$20–30) for the LLM arm; if unavailable, we will evaluate LLMs on a smaller sample. The scoped fallback is evaluation-focused rather than a different project: if weak-label mining proves too noisy, we reduce the number of acronym types but keep only high-confidence examples; if fine-tuning yields no meaningful signal after sanity checks, we still deliver the benchmark, baselines, LLM comparison, and error analysis as a complete empirical contribution.

Selected papers

- [1] Amir Pouran Ben Veyseh, Franck Dernoncourt, Quan Hung Tran, and Thien Huu Nguyen. *What Does This Acronym Mean? Introducing a New Dataset for Acronym Identification and Disambiguation*. COLING 2020. <https://doi.org/10.48550/arXiv.2010.14678>
- [2] Yaakov HaCohen-Kerner, Ariel Kass, and Ariel Peretz. *HAADS: A Hebrew Aramaic Abbreviation Disambiguation System*. Journal of the American Society for Information Science and Technology, 2010. Hebrew/Aramaic precedent; included as task-specific context alongside the post-2018 NLP anchors. <https://doi.org/10.1002/asi.21367>
- [3] Shaked Shmidko, Avi Shmidko, Moshe Koppel, and Chaim J. Cohen. *DictaBERT: A State-of-the-Art BERT Suite for Modern Hebrew*. arXiv preprint, 2023. <https://arxiv.org/abs/2308.16645>